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# Lesson 42 Recording and earthquake

记录地震 被动语态 强调句

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earthquake / ˈɜːθkweɪk / n. 地震
ninepin / ˈnaɪnˌpɪn / n. 九柱戏中的木柱
delicate / ˈdelɪkət / adj. 灵感的
penholder / ˈpenˌhəʊldə(r) / n. 笔杆
drum / drʌm / n. 鼓状物
bluebottle / ˈbluːbɒtl / n. 绿头苍蝇
graphic / ˈgræfɪk / adj. 图示的
transverse / ˈtrænzvɜːs / adj. 横向的
slumber / ˈslʌmbə(r) / v. 睡眠
rigid / ˈrɪdʒɪd / adj. 坚硬的
seismometer / saɪzˈmɒmɪtə(r) / n. 地震仪
legibly / ˈledʒəbli / adv. 字迹清楚地
wriggle / ˈrɪgl / v. 扭动
graph / grɑːf / n. 图表
longitudinal / ˌlɒŋgɪˈtjuːdən(ə)l / adj. 纵向的

## 记录地震的需求 (笔记最后修改于:2026-04-11 05:31:32)

※ An earthquake comes like a thief in the night, without warning.

『like a thief in the night 不期而至/不知不觉；偷偷摸摸（习语）secretly or unexpectedly and without being seen』

- The years, **like a thief in the night**, have stolen what you were.
- Innocence slipped away from me **like a thief in the night**.
- The sudden market crash comes **like a thief in the night**, surprising everyone.突然的市场崩盘不期而至，让所有人都感到意外。
- Illness can sometimes come **like a thief in the night**, rapidly changing one's life.疾病有时会像夜间的盗贼一样到来，迅速改变一个人的生活。
- The unexpected heavy snowstorm came **like a thief in the night**, paralyzing the city.这场突如其来的大暴雪不期而至，使城市陷入瘫痪。

『without warning suddenly and with no signs that it was going to happen, used about bad or dangerous things （贬义） 突然，突如其来地;没有警告/毫无预兆』

- He blazed up **without warning**.

- Then, **without warning,** her husband left her.
- The system failed **without warning**, causing massive data loss.系统在毫无预兆的情况下失败了，导致大规模数据丢失。
- The animal attacked the hunter **without warning**.那只动物毫无预兆地袭击了猎人。

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※ It was necessary, therefore, to invent instruments that neither slumbered nor slept.

『 **therefore** adv.因此/所以(不特别强调"因果"关系，多做插入语，用于补充说明) 』

- There is still much to discuss. We shall, **therefore**, return to this item at our next meeting.
- Progress so far has been very good. We are, **therefore**, confident that the work will be completed on time.
- The road was slick with ice; **therefore**, we drove cautiously.道路上结了一层冰；因此，我们小心翼翼地开车。
- She missed the deadline for the application; **therefore**, her request was denied.她错过了申请截止日期；因此，她的请求被拒绝了。

『 **instrument** 1. 乐器 』

- We have an old **musical instrument**.

『 **instrument** 2. 仪器，设备 』

- His telescopic observations are justly immortal; they aroused great interest at the time, they had important theoretical consequences, and they provided a striking demonstration of the potentialities hidden in **instruments** and apparatus.

『 **slumber** v. (fml.) sleep, esp. peacefully and comfortably 睡眠（尤指睡得安稳而舒服） 』

- The baby was **slumbering** peacefully.

『 **neither slumbered nor slept** 既不打盹也不睡觉/时刻保持警醒（同义词连用，表强调，如："深仇大恨、美丽动人、拖拖拉拉..."） 』

- The security system was designed to neither **slumber nor sleep**, monitoring 24/7.该安全系统被设计成时刻保持警醒，进行24/7的监控。
- In times of crisis, the emergency team must neither **slumber nor sleep** until the threat is neutralized.在危机时期，应急小组必须时刻保持警醒，直到威胁被消除。

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※ Some devices were quite simple.

『 **quite simple** 相当简单 』

- The instructions for assembling the furniture were **quite simple**.组装家具的说明相当简单。
- The solution to the complex problem turned out to be **quite simple**.这个复杂问题的解决方案结果是相当简单。

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※ One, for instance, consisted of rods of various lengths and thicknesses which would stand up on end like ninepins.

『**on end** 1. upright 直立，竖着』

- He placed the box **on end** and sat on it.
- The reader's hair stands **on end** when he reads in the final pages of the novel ...

『**on end** 2. continuously 连续地』

- They argued for two hours **on end**.
- Some plays are so successful that they run for years **on end**.

『**ninepins** n. [U] a game in which you roll a ball at nine bottle-shaped objects to try to hit them so that they fall 九柱戏』

- The pillars were placed close together, standing up on end like **ninepins**.柱子被紧密地放置在一起，像九柱戏的木瓶一样立着。
- to fall down or become ill / sick in great numbers 大量倒下；大批病倒

『**consisted of** 由...组成/构成』

- The original menu only consisted of five items.最初的菜单只由五道菜组成。
- The exhibit hall mainly consisted of interactive displays and video screens.这个展厅主要由互动展示和视频屏幕构成。

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※ When a shock came, it shook the rigid table upon which these stood.

『**upon which (Formal Structure)** 在其上（正式结构）』

- The foundation is the base **upon which** the entire structure rests.地基是整个结构立在其上的基础。
- That decision was the turning point **upon which** his whole career hinged.那个决定是他的整个职业生涯所依赖的转折点。

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※ If it were gentle, only the more unstable rods fell. If it were severe, they all fell. 主真从虚（混合使用）教程视频第19分处

『**If it were gentle/severe** 如果是轻微的/严重的（虚拟语气）』

- If it **were** I, I would choose the more challenging option.如果是我，我会选择更具挑战性的选项。

> Demographers expect it to take decades for China's birth rate to recover — by which time China **may have become** the first country in history to **have grown** old before growing rich. (The Economist)

※ Thus the rods, by falling, and by the direction in which they fell, recorded for the slumbering scientist the strength of a shock that was too weak to waken him, and the direction from which it came.

『 **in ... direction** 向...方向 』

- Tony glanced **in** her **direction** and their eyes met.
- The girls pointed **in** the opposite **direction**.

『 **from ... direction** 从...方向 』

- There was a loud scream **from the direction of** the children's pool.

| | |
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| **in ... direction** 向...方向 | They decided to drive **in** a northerly **direction** to reach the coast.<br>他们决定向北行驶以到达海岸。 |
| **from ... direction**<br>从...方向 | A cold wind was blowing **from** the **direction** of the mountains. 一阵冷风从大山的方向吹来。 |

分析记录原理 (笔记最后修改于:2026-04-13 06:06:02)

※ But instruments far more delicate than that were needed if any really serious advance was to be made.

『 **serious** adj. [only before noun] infml. used to emphasize that you are talking about a large amount of something 大量的；重大的；非常多的 』

- The President was in **serious** trouble. 大麻烦
- It's time you made some **serious** money. 赚大钱

『 **be to do** § (fml.) 1、打算，想要 』

- I'**m to see** her tomorrow.
- The old farmhouse the family lived in **was to be** replaced by a new $100,000 home.

『 **be to do** (fml.) 2、命令 』

- You'**re to be** back by 10 o'clock. No one is to leave the building.
- You **are to wait** here in this room until I return.

『 **be to do** (fml.)3、能够 』

- How **am** I **to pay** such a debt?
- We Chinese people **are** not **to be bullied.**
- The contemporary phenomenon of car worship **is to be explained** not least by the sense of independence and freedom that ownership entails.

『 **be to do** (fml.)4、应该 』

- Such men **are to be pitied** rather than despised.
- As soon as he had hung up, he went over to the workmen and told them that if a policeman ordered them to go away, they **were not to take** him seriously.

> Some books **are to be tasted,** others **to be swallowed**, and some few t**o be chewed** and **digested**. — Francis Bacon

※ The ideal to be aimed at was to devise an instrument that could record with a pen on paper, the movements of the ground or of the table as the quake passed by.

『 **be aimed at ...** 1. 目的是... 』

- an energy program **aimed** at reducing our dependence on fossil fuels
- The regulations **are aimed at** the prevention of accidents at work.
- In education, the ideal to **be aimed at** is fostering critical thinking, not just memorization.在教育中，要达到的理想目标是培养批判性思维，而不仅仅是记忆。

『 **be aimed at ...** 2. 针对... 』

- The criticism **was not aimed at** you.
- The book **is aimed at** people with no specialized knowledge.

『 **quake** n. (infml.) earthquake 地震（通俗说法） 』

- The epicentre of **the quake** was near the town of Potenza. 地震震中在波坦察城附近。
- They recommended that the viaduct be replaced rather than **quake**-proofed. 他们建议更换这座高架桥，而不要对其做抗震处理。
- **The quake** was the biggest in 140 years. 这是140年来最大的一场地震。
- I've never felt **a quake** that strong, you know. 你知道，我从来没有遇到过如此强的地震。

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※ While I write my pen moves, but the paper keeps still.

| While I write | my pen moves | , but the paper keeps still. |
| --- | --- | --- |
| 时间状语从句 | 时间状语主句 | 并列句（主系表） |

※ With practice, no doubt, I could in time learn to write by holding the pen still while the paper moved.

『 **no doubt** 1. used for emphasizing that sth. seems certain or very likely 无疑，毫无疑问 』

- **No doubt** she'll tell me everything when she's ready.
- **No doubt** I had blundered again. 不用说我又犯了个蠢错。
- There is no **doubt about** her suitability for the job. 毫无疑问，她适合做这个工作。

『 **no doubt** 2. used for showing that you accept sth. is probably true, but this does not change your opinion 的确，诚然 』

- **No doubt** she's very nice, **but** I just don't like her.
- **No doubt** many will regard these as harsh words, **but** regrettably they are true.

- **No doubt** the plan has some merits, **but** the cost is simply too high. 的确，这个计划有一些优点，但成本实在太高了。

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| **no doubt ..., but ...** 诚然...，但是... | **No doubt** she is very talented, **but** she still needs to work on her team communication skills. 诚然她非常有才华，但她仍需要在团队沟通技巧上下一番功夫。 |

『 **in time** 1. before the time by which it is necessary for sth. to be done 及时 』

- The Titanic turned **just in time**, narrowly missing the immense wall of ice which rose over 100 feet out of the water beside her.
- It threatened to surround him completely, but Tazieff managed to escape **just in time**.
- I arrived **just in time** for my flight to Hawaii. 我正好及时赶上飞往夏威夷的航班。

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| **to be done** （主语）有待被做；要做 | There is still a lot of work **to be done** before the website goes live. 在网站上线之前，还有很多工作有待完成。 |
| **be in time to do sth.** 及时做某事；来得及做某事 | • Will I **be in time to** catch the train?<br>• We ran to the station and were just **in time to catch** the last train. 我们跑向车站，刚好赶上了最后一班火车。 |
| **be in time for sth.** 对某事来说是及时的 | • They hope to finish **in time for** a trade summit between America and the EU on May 28th.<br>• The following day, however, the doctor consoled him by telling him that his chances of being able to leave hospital **in time for** New Year celebrations were good. |

『 **in time** 2. after a certain period of time, and esp. after a gradual process of change or development 逐渐地，慢慢地 』

- **In time**, all Sam's money was paid back in this way.
- **In time,** it became an accepted fact that the Cox brothers employed a conscientious ghost that did most of their work for them.
- I'm sure you'll grow to like her **in time**. 我肯定你慢慢就会喜欢她了。
- If you remain patient, you will see results **in time**.如果你保持耐心，最终你会看到结果。

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※ That sounds a silly suggestion, but that was precisely the idea adopted in some of the early instruments (seismometers) for recording earthquake waves.

『 **adopt** v. take over and have or use sth. as one's own 采纳，采取，采用 』

- The system has worked so well, that it **has been adopted in** other countries like Denmark, Norway, Finland, and New Zealand.

- For this reason there is no risk to the customer in the practice, **adopted by** some banks, of printing the customer's name on his cheques.
- The final design was precisely the idea **adopted** from the initial sketch.最终的设计恰恰是被采纳的来自最初草图的想法。

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| **adopt a name** 采用/改用一个名字 | After moving to the UK, he decided to **adopt** an English **name** to make it easier for his colleagues. 移居英国后，他决定改用一个英文名，以便同事称呼。 |
| **adopt a custom** 采纳/遵循一种习俗 | Many immigrants **adopt** the local **customs** while still preserving their own traditions. 许多移民在保留自己传统的同时，也会遵循当地的习俗。 |
| **adopt an idea** 采纳一个想法/主意 | The committee liked the proposal and voted to **adopt the idea** for the upcoming festival. 委员会很喜欢这个提议，并投票决定在即将到来的节日中采纳这个想法。 |
| **adopt a style of dress** 采取/穿某种风格的衣服 | She chose to **adopt** a more formal **style of dress** after being promoted to executive. 在晋升为高管后，她选择采取一种更为正式的着装风格。 |

※ But when table, penholder and paper are all moving, how is it possible to write legibly?

『 **penholder** n. a holder or handle for a pen point 笔杆 』

- A tapered point of a pen, designed to be inserted into a **penholder** or fountain pen. 如果您在笔杆上插上金尖就成了金笔了。
- Analyzed the structure characteristics of the tip and **penholder** and molding technology, introduced the injection mold structure characteristics. 分析了笔尖和笔杆的结构特点与成型工艺，介绍了该类零件的注射模结构特点。

『 **legible** adj. (of print or handwriting) clear enough to be read easily（印刷或字迹）清楚的，易读的 』

- The inscription was still **legible**.
- Ability to provide **legible** communication. 能够提供清晰的交流。

『 **legibly** adv.清楚地 』

- Please write more **legibly**.
- The doctor's prescription was difficult to read, but the nurse managed to write it **legibly** for the pharmacist.医生的处方难以辨认，但护士设法为药剂师清晰地书写了出来。

『 **illegible** adj. difficult or impossible to read; not legible 字迹模糊难以辨认的 』

- His handwriting is **illegible**. ⇒ He has a **doctor's handwriting**.
- His handwriting is neat. ⇒ He has **a teacher's handwriting**.

※ The key to a solution of that problem lay in an everyday observation.

『 **the key to...** ...的关键 』

- Such extreme cases must probably be taken with a grain of salt, but they do underline the general principle that the relationship between elephant and mahout is **the key to** successful training.
- Understanding the user experience is **the key to** a solution of that problem.理解用户体验是解决那个问题的关键。

『 **lie in ...** 在于...;位于...之中 』

- The reason **lies in** its gravity.
- The breakthrough in the experiment **lay in** an unexpected observation made by a junior assistant.实验的突破在于一位初级助理做出的一个意想不到的观察。
- The strength of the team **lies in** its ability to collaborate under pressure. 这支团队的力量在于其在压力下协作的能力。

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| **the key to ... lies in ...** ...的关键在于... | • **The key to** success **lies in** diligence.<br>• **The key to** successful language learning **lies in** consistent daily practice. 语言学习成功的关键在于持之以恒的每日练习。 |

※ Why does a person standing in a bus or train tend to fall when a sudden start is made?

『 **tend to** 倾向于...；往往... 』

- Morals **tend to** be deeply ingrained. 道德观念往往是根深蒂固的。
- Young trees **tend to** fall in strong winds if their roots are not deep enough.如果根不够深，幼树在强风中倾向于跌倒。
- The drugs **tend to** cause drowsiness. 这些药常常使人昏昏欲睡。

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※ It is because his feet move on , but his head stays still.

『 **move on** 向前移动;继续 』

- Can we **move on** to the next item on the agenda? 我们可以接着讨论下一项议程吗?
- Let's **move on** to the prickly subject of taxation reform. 咱们继续讨论下一项税制改革这个棘手的问题吧。
- Go, if you must **move on** alone. 去吧，如果你必须独自上路。

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※ A simple experiment will help us a little further.

『 **a little further** 更进一步 』

- Collecting more data will help us **a little further** in confirming the hypothesis.收集更多数据将帮助我们在确认假设方面更进一步。
- As he moved the car **a little further** forward, the rope broke. 当他把车再往前挪一点时，绳子断了。
- They carry it **a little further** than that. 他们比普通人看的更远一些。
- Let me explain this **a little further**. 让我更进一步来解释一下这个问题。

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※ Tie a heavy weight at the end of a long piece of string.

『**string** n. [U & C] a strong thread made of several threads twisted together, used for tying or fastening things 线绳』

- I need a piece of **string** to tie this package. 不可数
- The piece segues into his solo with the **strings**. 可数
- He wrapped the package in brown paper and tied it with **string**. 他用棕色包装纸把包裹包好，又用细绳捆上。

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※ With the hand held high in the air, hold the string so that the weight nearly touches the ground.

『**hold** v. to put a part of your body into a particular position 举起(独立主格里的hold)』

- She was **holding** her hand to her heart.

『**high** (物理意义的"高") / **highly** 抽象的高（高度）』

- He climbed **high** up the mountain.
- It is **highly** confidential.
- The birds were flying **high** in the sky. (物理位置)
- He is a **highly** respected educator in his field. (抽象程度：高度地)

| | |
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| **deep**<br>**deeply** | • He pushed the stick **deep** into the mud.<br>• He was **deeply** moved.<br>• They dug **deep** into the ground to find water. (物理深度)<br>• I was **deeply** moved by the student's progress. (抽象感情：深刻地) |
| **wide**<br>**widely** | • Open your mouth **wide**.<br>• He seems to be **widely** known here.<br>• Open your mouth **wide** so the dentist can see. (物理宽度/程度)<br>• English is **widely** used as a second language. (抽象范围：广泛地) |

| | |
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| **sharp**<br>**sharply** | • We're meeting at one-thirty **sharp**.<br>• turn **sharp** left / right<br>• Interest rates have fallen **sharply** over the last few weeks.<br>• The government has been **sharply** criticized for human rights violations.<br>• Please turn **sharp** left at the next intersection. (物理/直接：突然地)<br>• Stock prices fell **sharply** after the news. (抽象变化：剧烈地) |
| **low**<br>adj./adv.<br>**lowly** adj. | • The plane flew **low**.<br>• We were, however, worried about our nearest neighbours, whose farm was **low** lying and who were newcomers to the district.<br>• a **lowly** servant adj.地位低<br>• If we were able to discover **lowly** forms of life like bacteria on another planet, it would completely change our view of ourselves.<br>• The plane was flying very **low** over the trees. (物理高度)<br>• He started his career in a **lowly** position at the firm. (抽象地位：卑微的/地位低的) |

地震记录的机器 (笔记最后修改于:2026-04-13 09:34:27) 5

※ Now move the hand to and fro and around but not up and down.

『 **to and fro** if sb. or sth. moves to and fro, they move in one direction and then back again 来 来回回地 』

- The swing rocked **to and fro**.
- I was disturbed by all the people walking **to and fro** outside the office.
- The boat rocked **to and fro** on the choppy water.船在波涛汹涌的水面上来回摇晃。

| | |
|---|---|
| **back and forth** 来来回回地<br>(moving first in one direction and then in the opposite one) | • She swayed gently **back and forth** to the music.<br>• He paced **back and forth**. 他来回踱步。 |
| **from side to side** 来来回回地<br>(from left to right and from right to left) | • The curtains were swinging from side to side in the breeze.<br>• The Cutty Sark rolled **from side to side** and it became impossible to steer her.<br>• But as the paper moved **from side to side** under the pen point, its movement would be recorded in ink upon its surface. |

『 **up and down** if sth. moves up and down, it moves towards a higher position and then back to a lower position 上上下下地 』

- The boat was moving **up and down** on the sea.
- The earth trembled and the house seemed to be moving **up and down**.
- The crane can move forward and backward, but not **up and down**.起重机可以向前和向后移动，但不能上下移动。

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※ It will be found that the weight moves but slightly or not at all.

『 **moves slightly** 稍微一动 』

- The needle on the gauge **moved but slightly**, indicating a minor change.仪表上的指针只移动了一点点，表明变化很小。
- You can easily see all their bones' shapes as long as they **move slightly**! 肩部的骨头，只要稍稍一动就可以看到所有的骨头形状！

---

※ Imagine a pen attached to the weight in such a way that its point rests upon a piece of paper on the floor.

『 **Imagine** v.想象，设想 』

- **Imagine** their dismay when they found a beautifully-cooked wallet and notes turned to ash!
- **Imagine** loving your enemy and doing good to those who hated you. 想像着爱你的敌人，为仇人做好事。
- One can **imagine** how regretful he was. 可以想像他是多么懊悔。

『 **be attached to ...** 1. 与...连在一起 』

- Lined up against the wall, there were long thin wires **attached to** metal spheres.
- The warning label was securely **attached to** the product.警告标签牢固地连接到产品上。
- A trailer **was attached to** the rear of the truck. 卡车后面挂了一辆拖车。

『 **be attached to ...** 2. 依恋... 』

- She **was** sentimentally **attached to** this house, for even though it was far too big for her needs, she persisted in living there long after her husband's death.
- She **is** very **attached to** her family and friends. 她非常喜欢她的家人和朋友。
- I've never seen two people so **attached to** each other. 我从未见过两个人如此形影不离。

『 **rest on / upon** sth. 1. 靠在...上面 』

- She stood three feet high and her hands **rested on** her hip.
- The weary traveler finally allowed his head to **rest upon** the soft pillow.疲惫的旅行者终于允许他的头停留在柔软的枕头上。

『 **rest on / upon sth.** 2. 基于... 』

- His fame **rests** more **on** his plays than on his novels.
- Does the economy **rest on** too narrow a base? 这种经济的基础是否过于狭窄?

『**in such a way that...** 以至于...的方式』

- She arranged the lights **in such a way** that they highlighted the painting perfectly. 她以至于它们完美地突显出画作的方式布置了灯光。
- And **in such a way** that it became a weapon in the cold war. 这样一来，它会变成冷战中的一件武器。
- The world has been shaped by technology **in such a way** that it is no longer surprising. 世界已经被技术用这样一种方式定型了，这已经没什么好惊讶的了。

---

※ Imagine an earthquake shock shaking the floor, the paper, you and your hand.

『**imagine sb. doing sth.** 想象某人正在做某事』

- I can easily **imagine** him **becoming** a successful entrepreneur in the future. 我完全可以想象到他未来成为一名成功企业家的样子。
- She could **imagine** dark-robed figures **moving** silently along the stone corridors.
- **Imagine** you and me **going** to outer space. 设想一下你和我将去到外太空。

---

※ In the midst of all this movement, the weight and the pen would be still.

『**in the midst of ...** 1. while sth. is happening or being done 在...发生期间』

- a country **in the midst of** a recession
- She discovered it **in the midst of** sorting out her father's things.
- She alone remained calm **in the midst of** all the confusion.

『**in the midst of ...** 2. in the middle of a place or a group of things or people 在...之间(地点)』

- I suddenly realized I was standing **in the midst of the crowd**, but I couldn't see my friends anywhere. 我突然意识到自己正站在人群之中，却到处也看不到我的朋友。
- He described it as 'a very agreeable situation located within two small hills **in the midst of** which flowed a great river'.
- The small cottage sat peacefully **in the midst of** a dense pine forest. 那座小木屋静静地坐落在茂密的松树林之中。
- **In the midst of** the chaos, she remained surprisingly calm. 在混乱之中，她出奇地保持着冷静。

---

※ But as the paper moved from side to side under the pen point, its movement would be recorded in ink upon its surface.

『**in ink** 用墨水写(钢笔)』

- Please write **in ink**, not **in pencil**.
- The data from the sensor would be recorded **in ink** upon the chart. 来自传感器的数据会以墨水的形式被记录在图表上。
- The letter was written **in ink**. 这封信是用钢笔写的。

※ It was upon this principle that the first instruments were made, but the paper was wrapped round a drum which rotated slowly.

『on / upon ... principle 依照...理论或原则』

- The country is run **on socialist principles**.
- The project worked **on the principle** that each person's experience was equally valuable.
- It **was upon this principle** that the new engine was designed.正是基于这个原理，新引擎被设计了出来。

『in principle 1. 原则上（但尚未考虑细节）』

- Politicians agree, **in principle**, so America and the EU have been trying to reach a deal which would eliminate the need to double-test many products.
- **In principle**, this makes sense. 原则上讲得通。
- Is there a difference **in principle**? 有没有原则上的区别呢?

『in principle 2. 理论上（但尚未发生）』

- **In principle**, the new software should make the accounting system a lot simpler.
- Even assuming this to be **in principle** possible, it will not be achieved soon.
- **In principle**, it should not be too hard. 原理上，这应该不会太难。

『wrap sth. round / around sb. / sth. 将某物缠绕/包裹在某人或某物上』

- Ella **wrapped** a thick coat **around** her shoulders.
- He **wrapped** a bandage **around** my injured wrist.
- She **wrapped** a warm scarf **around** her neck before heading out into the snow. 出门踏入雪地前，她把一条暖和的围巾围在脖子上。

『drum n. sth. that looks like a drum, esp. part of a machine 鼓状物，鼓状零件』

- The mixture flows to a revolving **drum** where the water is filtered out. 混合剂流过旋转着的滚筒时水分便从中滤出。
- The **oil drum** leaks. 油桶漏了。
- **Drum brake** with large clearance between the feet or oil. 制动蹄与制动鼓之间的间隙过大或蹄油。
- The electrical cable was neatly wrapped round **a drum** before shipping.电缆在运输前被整齐地缠绕在一个鼓筒周围。

---

※ As long as all was still, the pen drew a straight line, but while the drum was being shaken, the line that the pen was drawing wriggled from side to side.

『as long as 在...的时候，只要...就』

- **As long as** you're happy, that's all that matters. 只要你幸福，这就是最重要的事情。

- **As long as** all was still with the patient, the doctor decided to wait. 只要病人的情况一切保持静止，医生决定等待。
- **As long as** you have your health, nothing else matters. 只要身体健康，其他任何事都无关紧要。

| | |
|---|---|
| **so long as**<br>只要... | **So long as** there is a demand for these drugs, the financial incentive for drug dealers will be there.<br>只要对这些毒品有需求，也就存在着对贩毒者的经济诱因。 |

※ The apparatus thus described, however, records only the horizontal component of the wave movement, which is, in fact, much more complicated.

『 **apparatus** n. [U] = equipment 仪器，设备 』

- With experience, and with improved **apparatus**, it is now possible not only to locate a shoal but to tell if it is herring, cod, or other well-known fish, by the pattern of its echo.
- His telescopic observations are justly immortal; they aroused great interest at the time, they had important theoretical consequences, and they provided a striking demonstration of the potentialities hidden in instruments and **apparatus**.

『 **thus** adv. in this way, like this 以此方式，如此，这样 』

- calculate the area of the triangle **thus** formed 计算**这样**形成的三角形的面积
- Hold the wheel in both hands, **thus**. 用双手握住方向盘，像**这样**。
- The process **thus described** requires specialized tools. 如此描述的过程需要专门的工具。

---

※ If we could actually see the path described by a particle, such as a sand grain in the rock, it would be more like that of a bluebottle buzzing round the room; it would be up and down, to and fro and from side to side.

『 **path** n. line along which sb./sth. moves （人或事物移动的）路线，轨道 』

- the moon's **path** round the earth
- the **path** of a tornado
- Scientists used advanced imaging to trace **the path** described by a photon. 科学家们使用先进的成像技术来追踪一个光子所描述的路径。

『 **grain** n. [C] a single very small piece of a substance such as sand or salt 小颗粒 』

- The recipe only calls for **a grain of salt** to enhance the flavor of the chocolate. 食谱只需要一粒盐来提升巧克力的风味。
- Even **a grain of sand** in your eye can cause significant discomfort. 哪怕是一粒沙子进到眼里，也会让你感到非常不适。
- There wasn't **a grain of sugar** left in the jar after she finished baking the cake. 她做完蛋糕后，罐子里连一粒糖都不剩了。

『 **bluebottle** n. large buzzing fly with a blue body 青蝇 』

- We could see a **bluebottle** on the rock. 我们能看见一只绿头苍蝇在岩石上。
- A **bluebottle** buzzed around the garbage bin. 一只蓝蝇在垃圾桶周围嗡嗡作响。
- The picnic was ruined by the presence of several **bluebottles**. 几只蓝蝇的出现毁了这次野餐。

『 **buzz** v. to make a continuous sound, like the sound of a bee 发出嗡嗡声 』

- The city has a real **buzz**. 这个城市真热闹。
- **A low buzz of** admiration swept through the assemblage. 一阵表示敬佩的低沉的赞赏声在聚集的人群中传开。
- **A buzz of** excitement filled the courtroom as the defendant was led in. 被告被带进来时，法庭内充满激动的嘈杂声。

---

※ Instruments have been devised and can be so placed that all three elements can be recorded in different graphs.

『 **element** n. one part or feature of a whole system, plan, piece of work etc., esp. one that is basic or important 要素，重要部分 』

- Honesty is a vital **element** of her success.
- Besides ability, the other essential **element** in political success is luck.
- There is **an element of** subjectivity in her criticism. 她的批评有主观因素。

『 **graph** n. a picture that uses lines or curves to show the relationship between numbers or measurements that change 图，图表，曲线图 』

- A bar **graph** will show you the busy and quiet times during the week. 柱状图会显示一周中繁忙和平静的时间段。
- I first saw a **graph** similar to this one in a software magazine article. 我在一篇软件杂志的文章中第一次看到与此类似的图。
- We plotted about eight points on the **graph**. 我们在图表上绘出了大约有八个点的曲线。

**波动传导方式** (笔记最后修改于:2026-04-13 10:29:01) 3

※ When the instrument is situated at more than 700 miles from the earthquake centre, the graphic record shows three waves arriving one after the other at short intervals.

『 **one after the other = one after another** used to talk about a series of similar things or events 一个接一个地 』

- I'll eat chocolates **one after the other** until the box is finished.
- Small businesses have been collapsing **one after another**.

『 **interval** n. （时间上的）间隔；（空间上的）间距 』

- at weekly / 20 minute **intervals**
- The trains run at hourly **intervals**.

- The trees were planted at 20 meter **intervals**.
- But suppose the earth could be covered with sensors spaced one foot apart, rising **at** one-foot **intervals** all the way to the top of the atmosphere.
- Indeed, it may well be that those whose work is their pleasure are those who most need the means of banishing it **at intervals** from their minds.
- There was a short **interval** of peace before the noise started again. 在噪音再次开始前，有一段短暂的宁静。
- The security camera took pictures **at short intervals** throughout the night. 这台安全摄像机整晚都以短间隔拍照。

| | |
|---|---|
| **at ... intervals** 每隔...（时间或空间） | The lamp posts were placed **at** regular **intervals** along the road. 路灯沿路每隔一段距离设置一盏。 |

※ The first records the arrival of longitudinal vibrations.

『 **records the arrival of** 记录...的到达 』

- The first records the arrival of longitudinal vibrations. 第一个记录了纵向振动的到达。
- The sensor records the arrival of the first seismic activity. 传感器记录了首次地震活动的到达。

『 **longitudinal vibrations** 纵向振动 』

- The results show that **longitudinal vibration of drill** string is relative to rotary speed and weight on bit (WOB). 实验结果表明，钻柱的纵向振动主要与转速和钻压有关。
- The expression format and application method of **longitudinal vibration** road spectrum were described. 建立了纵振动力学模型，阐述了纵振路面谱的表达形式及应用方法。
- In this paper, the regularity of **longitudinal vibration** was analyzed by using finite element method. 本文利用有限元方法分析了钻柱轴向振动规律。

| | |
|---|---|
| **transverse vibrations** 横向振动 | The second marks the arrival of transverse vibrations. 第二个标志着横向振动的到达。 |

※ The second marks the arrival of transverse vibrations which travel more slowly and arrive several minutes after the first.

『 **mark the arrival of** 标志着...的到达 』

- The loud whistle **marks the arrival of** the express train. 响亮的汽笛声标志着特快列车的到达。
- The snow **signalled** the arrival of a front, and a high-pressure area seemed to be settling in. 降雪标志了一个锋面的到来，一个高压区似乎正在进入本地。
- With **the arrival of spring**, the refugees will be moved again. 随着春天的到来，难民们将再次被转移。

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※ These two have travelled through the earth.

『through the earth 贯穿地球（传播）』

- I want to dig a hole all the way **through the earth**! 我想挖一个贯穿地球的洞。
- I wonder if I shall fall right **through the earth**! 我想知道我会不会正好穿过地球掉下去！
- The deep-sea currents have **travelled through** the ocean basins for millennia. 深海洋流穿过洋盆传播了数千年。

---

※ It was from the study of these that so much was learnt about the interior of the earth.

『interior n. the inner part or inside of something 内部，里面』

- The **interior** of the church was dark.
- a house with a classical **exterior** and a modern **interior**

---

※ The third, or main wave, is the slowest and has travelled round the earth through the surface rocks.

『or conj. (introducing a word or phrase that explains, or means the same as, another 表示一词语对另一词语的解释关系或为其同义词语) 即，亦即，或者说』

- a kilo, **or** two pounds
- geology, **or** the science of the earth's crust
- Cave exploration, **or** pot-holing, as it has come to be known, is a relatively new sport.

Play it by ear

随机应变；走着瞧

Idioms

## 课后练习题 (解题方法与答案需要在“ncego.com”笔记官网查看)

1. W:Did Linda ever finish that introductory chapter?M: I'm not sure. She's spent hours on end rewriting it.Q: What does the man imply about Linda?
(单选题)

A、 She rearranged the chapters of her book.
B、 She assured him that the chapter was finished.
C、 She worked on the chapter for quite a while.
D、 She wasn't sure how to end the book.

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